Supplementary Material: Precision Accounting for KV and Recurrent-State Caches in Hybrid LLM Serving

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TABLE I
ARCHITECTURE-DERIVED CONTINUOUS-MODEL PARAMETERS. ALL VALUES ARE BYTES. NO PARAMETER IS FITTED TO ALLOCATOR TOKEN COUNTS.

Model	A_f	A_q	G_{fp32}	G_{bf16}
2B	12,288	3,168	19,537,920	10,100,736
9B	16,384	4,224	26,050,560	13,467,648

I. Byte-exact parameter derivation

The attention term includes all GQA layers. At fp16, each attention layer stores K and V tensors with 512 elements per token and 2 bytes per element, for 2,048 bytes per layer and token. At int4, K and V each use 256 packed payload bytes and 8 metadata bytes. Their combined cost is $2 \times (256 + 8) = 528$ bytes per layer and token.

Each GDN layer contains convolution and temporal state. The convolution state remains bf16 in both configurations and uses 36,864 bytes. The temporal state uses 1,048,576 bytes at fp32 and 524,288 bytes at bf16. Consequently,

$$G_{\text{fp32}}^{\text{layer}} = 36{,}864 + 1{,}048{,}576 = 1{,}085{,}440, \qquad (1)$$

$$G_{\text{bf16}}^{\text{layer}} = 36{,}864 + 524{,}288 = 561{,}152. \qquad (2)$$

The complete layer state falls by 48.30%, not 50%.

The 2B model has six GQA layers and 18 GDN layers; the 9B model has eight and 24, respectively. Multiplying the per-layer values gives the parameters in Table I. The representative padded recurrent pages are 1,089,792 bytes at fp32 and 566,016 bytes at bf16. Padding therefore adds 4,352 and 4,864 bytes, respectively. The continuous model uses unpadded architecture bytes; the discrete allocator uses the padded layout.

II. Discrete allocator reconstruction

Let K be the GPU-block count, H the recurrent cache-group count, B the attention block size, and L the requested context. The allocator-equivalent slot count and allocator token capacity are

$$S_{\text{alloc}} = \frac{K}{H + \lceil L/B \rceil}, \qquad T_{\text{alloc}} = \lfloor L S_{\text{alloc}} \rfloor. \qquad (3)$$

For 2B/int4/4K with fp32 state, $K = 3287$, $H = 3$, and $B = 2064$. The denominator is $3 + \lceil 4096/2064 \rceil = 5$. Thus $S_{\text{alloc}} = 657.4$ and $T_{\text{alloc}} = 2{,}692{,}710$. With bf16 state, $K = 6330$ and $B = 1072$. The denominator is seven, $S_{\text{alloc}} = 904.286$, and $T_{\text{alloc}} = 3{,}703{,}954$.

Fractional slots are an allocator normalization, not fractional scheduler admissions. They also do not establish admission of the integer floor under prefill, decode, or latency constraints.

III. Full-matrix and temporal checks

The formal matrix contained 112 allocator cells and 52 matched fp32/bf16 core pairs. Every pair favored bf16 temporal state. The median capacity gain was 15.44%, with a range of 3.32–93.28%. The corrected continuous-model residual had a 2.3795% median absolute magnitude, a 2.7955% mean absolute magnitude, and a 13.2134% maximum absolute magnitude. Signed residuals spanned -5.1604% to $+13.2134\%$.

The temporal rerun repeated the frozen matrix. Because the allocator is deterministic for a fixed build and configuration, this was a reproducibility check rather than an independent statistical replicate. All cell directions agreed, and the maximum token difference was 1.42%.

IV. Dependence-aware GSM8K analysis

Each dataset seed samples 200 GSM8K items without replacement. Samples overlap across seeds, producing 1,800 seed-item draws but only 1,017 unique items. Pairwise seed overlap ranges from 19 to 39, with median 30. The primary estimand is the mean paired difference over the observed seed-item draws.

For each candidate, an intercept-only OLS model is fit to paired binary differences. The covariance uses two-way CR1 clustering by item and dataset seed. The reference distribution is Student-t with $\min(1017-1, 9-1) = 8$ degrees of freedom. A percentile bootstrap samples the 1,017 item-level means with equal item weight for 20,000 replicates.

Repeated items were not always deterministic across dataset seeds despite greedy decoding. The 2B full, state-only, KV-only, and joint allocations had 30, 27, 47, and 43 repeated items with inconsistent outcomes. Clustering accounts for their

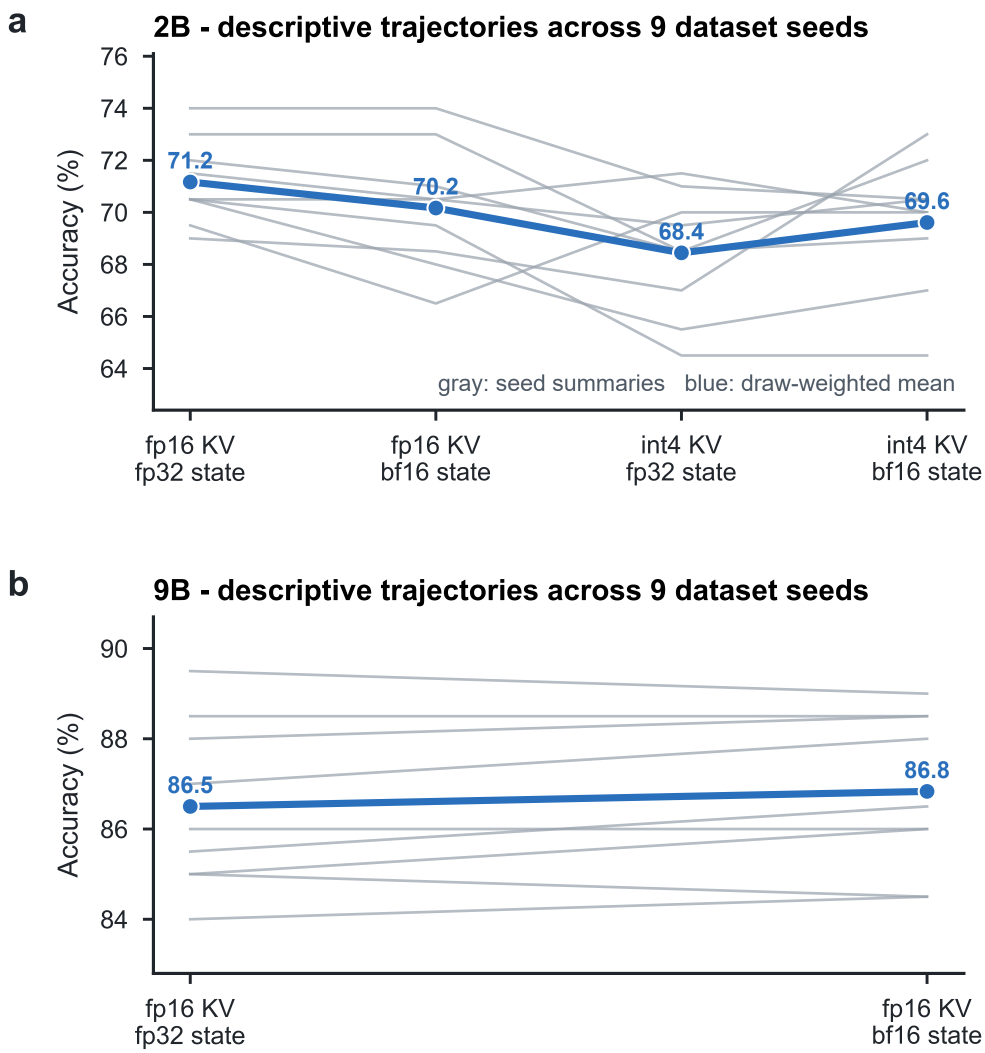

Fig. 1. **Descriptive GSM8K dataset-seed summaries.** Thin lines show nine seed summaries. The blue line is the draw-weighted mean. These lines are not independent replicates and are not the basis of the confidence intervals.

dependence but does not explain the behavior. To prevent reuse, the deprecated analysis that treated seed means as independent now exits with an error.

V. Additional Evidence Panels

VI. Artifact and Verification Manifest

The current submission does not claim a public artifact release. The repository-local verifier checks SHA-256 digests for 16 frozen sources, asserts headline values and all 52 paired capacity rows, and rejects obsolete capacity inputs and GSM8K inputs based on seed-level independence. It also checks all eight SVG/PDF pairs for nonempty vector output. Before producing the JSON and LaTeX tables, the selector-audit generator verifies all four decision sidecars.

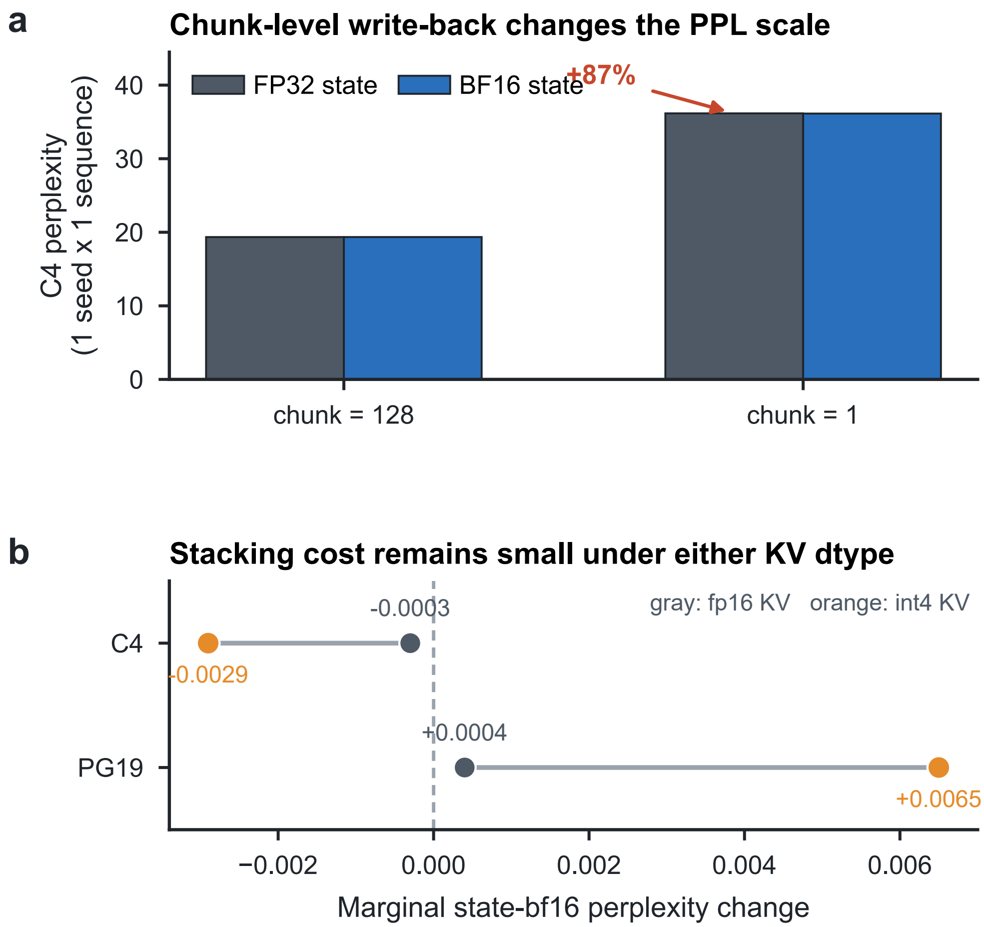

Fig. 2. **PPL harness boundary.** (a) Reducing chunk size from 128 to one changes the PPL scale by about 87%, so the harness is not equivalent to per-token kernel semantics. (b) State contrasts under fp16 and int4 KV have wide three-seed uncertainty and do not establish equivalence.

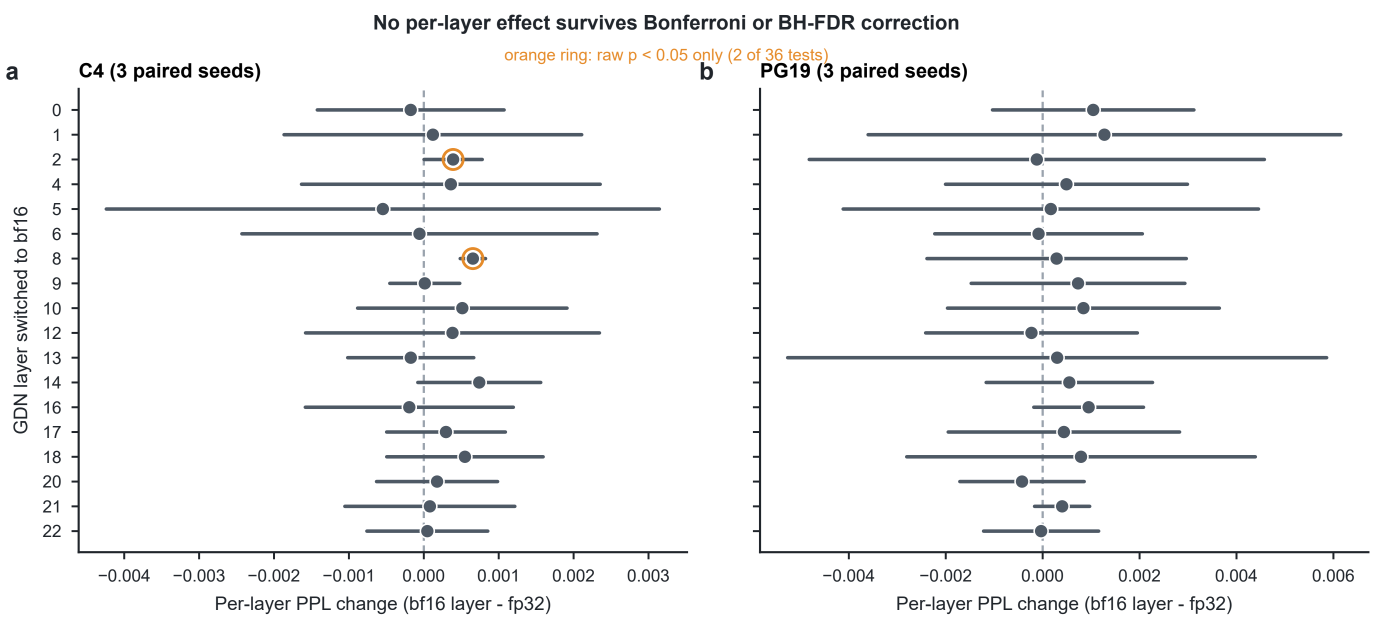

Fig. 3. **Exploratory per-layer sensitivity.** Two of 36 raw tests have $p < 0.05$. None survives Bonferroni or Benjamini–Hochberg correction. This screen does not establish that whole-state precision is optimal.

VII. Complete paired allocator matrix

Table II enumerates every fp32/bf16 pair in the formal 112-cell run. Model, KV data type, context length, and GPU-memory utilization uniquely identify each pair. The remaining eight cells probe a float16-state frontier and are not part of the 52-pair state comparison. The frozen source is `capacity-phase-formal-corrected.analysis.json`; its SHA-256 prefix is `879dc059579e`, with the full digest in the adjacent sidecar.

TABLE II: **Complete formal allocator matrix.** K is the allocated GPU-block count and T is total allocator token capacity. Gain is $100(T_{16}/T_{32} - 1)$; residual is the signed measured-to-continuous state-ratio error. The 52 rows come from attempt `capacity-phase-formal-20260811`; they are allocator measurements, not scheduler-admitted concurrency.

Model	KV	L	u	K_{32}	K_{16}	T_{32}	T_{16}	Gain	Resid.
2B	fp16	1,024	0.70	2,466	4,658	505,036	681,398	34.92%	-4.72%
2B	fp16	1,024	0.80	2,969	5,609	608,051	820,516	34.94%	-4.70%
2B	fp16	1,024	0.90	3,473	6,561	711,270	959,780	34.94%	-4.71%
2B	fp16	2,048	0.70	2,465	4,657	721,188	867,048	20.22%	-5.16%
2B	fp16	2,048	0.80	2,969	5,609	868,644	1,044,293	20.22%	-5.16%
2B	fp16	2,048	0.90	3,473	6,561	1,016,100	1,221,538	20.22%	-5.16%
2B	fp16	4,096	0.70	2,466	4,658	918,248	1,059,953	15.43%	-0.16%
2B	fp16	4,096	0.80	2,969	5,609	1,105,547	1,276,359	15.45%	-0.14%
2B	fp16	4,096	0.90	3,473	6,561	1,293,218	1,492,992	15.45%	-0.15%
2B	fp16	8,192	0.70	2,465	4,657	1,062,804	1,192,192	12.17%	+3.37%
2B	fp16	8,192	0.80	2,969	5,609	1,280,107	1,435,904	12.17%	+3.36%
2B	fp16	8,192	0.90	3,473	6,561	1,497,411	1,679,616	12.17%	+3.36%
2B	fp16	16,384	0.70	2,466	4,658	1,188,321	1,271,944	7.04%	+2.46%
2B	fp16	16,384	0.80	2,969	5,609	1,430,708	1,531,630	7.05%	+2.48%
2B	fp16	16,384	0.90	3,473	6,561	1,673,577	1,791,590	7.05%	+2.48%
2B	fp16	32,768	0.70	2,466	4,658	1,262,592	1,304,558	3.32%	+1.01%
2B	fp16	32,768	0.80	2,969	5,609	1,520,128	1,570,903	3.34%	+1.03%
2B	fp16	32,768	0.90	3,473	6,561	1,778,176	1,837,528	3.34%	+1.03%
2B	int4	1,024	0.70	2,515	4,843	643,840	1,239,808	92.56%	+12.80%
2B	int4	1,024	0.80	3,019	5,835	772,864	1,493,760	93.28%	+13.21%
2B	int4	1,024	0.90	3,545	6,826	907,520	1,747,456	92.55%	+12.79%
2B	int4	2,048	0.70	2,515	4,843	1,287,680	1,983,692	54.05%	-1.81%
2B	int4	2,048	0.80	3,030	5,835	1,551,360	2,390,016	54.06%	-1.80%
2B	int4	2,048	0.90	3,545	6,826	1,815,040	2,795,929	54.04%	-1.81%
2B	int4	4,096	0.70	2,515	4,843	2,060,288	2,833,846	37.55%	-2.38%
2B	int4	4,096	0.80	3,030	5,835	2,482,176	3,414,308	37.55%	-2.37%
2B	int4	4,096	0.90	3,545	6,826	2,904,064	3,994,185	37.54%	-2.38%
2B	int4	8,192	0.70	2,515	4,843	2,943,268	3,606,714	22.54%	-2.88%
2B	int4	8,192	0.80	3,030	5,835	3,545,965	4,345,483	22.55%	-2.88%
2B	int4	8,192	0.90	3,545	6,826	4,148,662	5,083,508	22.53%	-2.89%
2B	int4	16,384	0.70	2,515	4,822	3,745,978	4,158,086	11.00%	-3.66%
2B	int4	16,384	0.80	3,030	5,835	4,513,047	5,031,612	11.49%	-3.24%
2B	int4	16,384	0.90	3,545	6,805	5,280,116	5,868,058	11.14%	-3.55%
2B	int4	32,768	0.70	2,515	4,843	4,337,448	4,667,512	7.61%	-0.62%
2B	int4	32,768	0.80	3,030	5,835	5,225,633	5,623,567	7.62%	-0.62%
2B	int4	32,768	0.90	3,545	6,826	6,113,818	6,578,657	7.60%	-0.63%
9B	fp16	2,048	0.80	290	562	84,845	104,634	23.32%	-2.71%
9B	fp16	2,048	0.90	484	940	141,604	175,010	23.59%	-2.50%
9B	fp16	4,096	0.80	290	563	107,985	121,370	12.40%	-2.79%
9B	fp16	4,096	0.90	484	940	180,224	202,644	12.44%	-2.75%
9B	fp16	16,384	0.80	290	562	135,753	143,872	5.98%	+1.45%
9B	fp16	16,384	0.90	484	940	226,567	240,640	6.21%	+1.67%
9B	fp16	32,768	0.80	288	560	142,987	147,984	3.49%	+1.18%
9B	fp16	32,768	0.90	483	938	239,802	247,874	3.37%	+1.06%
9B	int4	2,048	0.80	287	567	146,944	232,243	58.05%	+0.74%
9B	int4	2,048	0.90	482	942	246,784	385,843	56.35%	-0.34%
9B	int4	4,096	0.80	288	559	235,929	327,094	38.64%	-1.60%
9B	int4	4,096	0.90	482	950	394,854	555,885	40.78%	-0.08%
9B	int4	16,384	0.80	288	567	428,962	488,933	13.98%	-1.08%
9B	int4	16,384	0.90	478	950	711,959	819,200	15.06%	-0.14%
9B	int4	32,768	0.80	286	564	493,244	528,032	7.05%	-1.14%
9B	int4	32,768	0.90	481	948	829,547	887,544	6.99%	-1.19%